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Introduction to Euclid's Geometry



Fruits and vegetables come in a variety of geometric shapes, sizes, and symmetry; a coconut, for instance, is a sphere, and after splitting it in two halves, one can see how the individual pieces are in a perfect hemispherical shape..



What's Inside

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- ✓ Caution Points
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Questions

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| TOPIC 1 |

INTRODUCTION

Learning Objectives

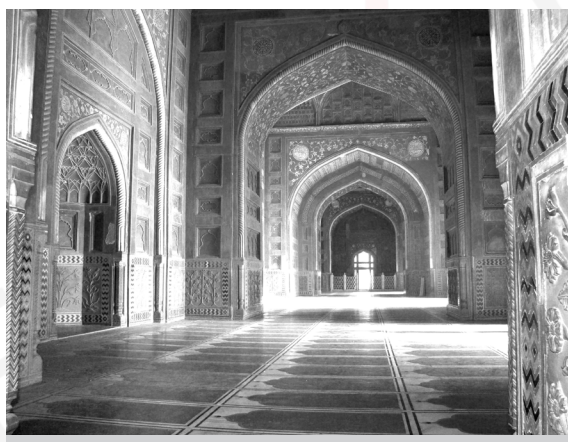
- Students will be learning about the great mathematician Euclid and his contributions in Mathematics.

Learning Outcomes

- Students will be able to understand the term geometry and its origin.

Real Life Application

Taj Mahal, one of the wonders of the world, is strikingly beautiful on the inside as it is on the outside. When you visit the Taj Mahal, you see many architectural designs on the wall of the monument. The Taj Mahal is a masterpiece of architecture. Have you ever thought how the architects made this beautiful monument? Euclidean geometry was the only one used for over two thousand years. It played a vital role in the design of the Taj Mahal.



Architects use Euclid's geometry to study and divide patterns, as well as draw designs. They rely on Euclidean geometric principles to create designs. The architectural design is based on geometric structures that develop out of the idea of transformations. The symmetry transformations are visible as design concepts throughout the history of architecture.

In our daily lives, we come across various buildings with varying shapes and figures that can only be explained by applying Euclid's geometry. Let's now study and learn more about the Euclid's geometry in detail.

The credit of introducing geometrical concepts goes to the Greek mathematician 'Euclid' who is famous with the name 'father of Geometry' and his work is also called 'Euclidean Geometry'. The word geometry comes from Greek words 'geo' which means 'Earth' and 'metrein' which means 'measure.'

Euclid's Definition

A solid has shape, size, position, and can be moved from one place to another. Its boundaries are called surface. They separate one part of the space from another and are said to have no thickness. The boundaries of the surface are curves or straight lines. These lines end in points.

Euclid summarised the properties of solid shape as definition and began his exposition by listing 23 definitions in Book 1 of the 'Elements'. Some of his fundamentals about solid shapes are as follows:

- (1) A point is that which has no part.
- (2) The edges of surface are lines.
- (3) A boundary is an extremity of anything.
- (4) A straight line is a line which lies evenly with the points on itself.
- (5) A plane surface is that which lies evenly with the straight lines on itself.
- (6) A figure is that which is contained by any boundary or boundaries.
- (7) A line is breadthless length.
- (8) Parallel straight lines are straight lines which, being in the same plane and being produced indefinitely in both directions, do not meet one another in either direction.

| TOPIC 2 |

EUCLID'S AXIOMS AND POSTULATES

Learning Objectives

- Students will be learning to know about the axioms and postulates of Euclid's geometry.

Learning Outcomes

- Students will be able to differentiate axioms and postulates and its applications in various geometrical concepts.

Real Life Application

Everything in the world of science needs proof. Without proof, the world seems very senseless. Remember, when solving equations in mathematics, we prove that the left-hand side is equal to the right-hand side? But there are some facts that we have been listening to and studying since our childhood that do not need to be proved.





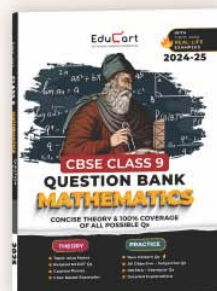
We have read and heard since our childhood that the Earth rotates 360 degrees every day. So, day and night are caused by the rotation of the earth on its axis. It is a fact that does not require any proof.

This is a basic application of axioms which can be seen in our day to day life. Let us study Euclid's axioms and postulates in detail



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OBJECTIVE Type Questions

[1 mark]

Multiple Choice Questions

1. Euclid stated that all right angles are equal to each other in the form of:

(a) an axiom (b) a definition
(c) a postulate (d) a proof

[NCERT Exemplar]

Ans. (c) a postulate

Explanation: "All right angles are equal to each other" is in accordance with Euclid's fourth postulate.

2. Which of the following statement is false?

(a) A straight line may be drawn from any one point to any other point.
(b) A terminated line cannot be produced indefinitely.
(c) A circle can be drawn with any center and any radius.
(d) All right angles are equals to one another.

[Diksha]

Ans. (b) A terminated line cannot be produced indefinitely.

Explanation: According to Euclid's second postulate, a terminated line can be produced indefinitely.

Assertion and Reason (A-R)

In these questions, a statement of Assertion (A) is followed by a statement of Reason (R).

Choose the correct option as:

- (a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).
(b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).
(c) Assertion (A) is true but reason (R) is false.
(d) Assertion (A) is false but reason (R) is true.

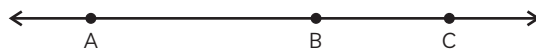
3. Assertion (A): If A, B and C are three points on a line, and B lies between A and C, then $AB + BC = AC$.

Reason (R): The things with one another are equal to one another.



Ans.(a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).

Explanation:



AC coincides with $AB + BC$.

Also, Euclid's Axiom (4) says that things which coincide with one another are equal to one another. So, it can be deduced that

$$AB + BC = AC$$

⚠ Caution

→ Assume that there is a unique line passing through two points.

CASE BASED Questions (CBQs)

[4 & 5 marks]

Read the following passages and answer the questions that follow:

4. Shreya was very fond of reading books. So, she went to the National Book fair at Pragati Maidan in New Delhi to get some story books. She decides to take 5 books from each store.



(A) Does Euclid's axioms apply in this?

- (a) Yes
- (b) No
- (c) Can't say anything
- (d) None of the above

(B) Which axiom is applied here?

- (a) First (b) Second
- (c) Third (d) Fourth

(C) What does the axiom state?

- (a) Things which are equal to the same thing are also equal to one another.
- (b) A straight line may be drawn from any point to any other point.
- (c) If equals are added to equals, then the wholes are equal.
- (d) The whole is greater than the part.
- (D) It is known that if equals are added to equals, then the wholes are equal, then Euclid's axiom that illustrates this statement is:
- (a) First Axiom (b) Second Axiom
- (c) Third Axiom (d) Fourth Axiom
- (E) If $z - 13 = 37$, then the value of z is:
- (a) 65 (b) 54
- (c) 50 (d) 44

Ans.(A) (a) Yes

(B)(b) *Second*

Explanation: Shreya buys 5 books from each store that means 5 books will be added to the total number of books.

(C)(c) *If equals are added to equals, then the wholes are equal.*

Explanation: If equals are added to equals, then the wholes are equal.

(D)(b) *Second Axiom*

Explanation: Euclid's second axiom: To produce (extend) a finite straight line continuously in a straight line: If equals are added to equals, then the wholes are equal (Addition property of equality).

(E)(c) 50

Explanation: $z - 13 = 37$

$$z - 13 + 13 = 37 + 13$$

$$z = 50$$

VERY SHORT ANSWER Type Questions (VSA)

[1 mark]

5. An equilateral triangle is a polygon made up of three line segments out of which two line segments are equal to the third one and all the angles are 60° each. Can you justify that all sides and all angles are equal in equilateral triangle? [Delhi Gov. QB 2022]

Ans. All the angles are 60° each equilateral triangle and two line segments are equal to third one, therefore all three sides of an equilateral triangle

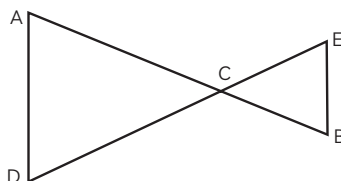


are equal (according to Euclid's axiom, things which are equal to the same thing are equal to one another).

SHORT ANSWER Type-I Questions (SA-I)

[2 marks]

6. In the given figure $AC = DC$, $CB = CE$, then show that $AB = DE$



[Delhi Gov. QB 2022]

Ans. We have $AC = DC$

...(i) [Given]

And $CB = CE$

...(ii) [Given]

Now, by axiom 2, if equals are added to equals, the wholes are equal.

Adding eqs. (i) and eqs. (ii).

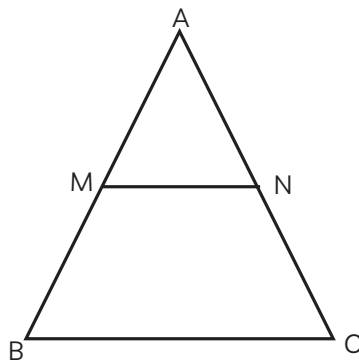
We get $AC + CB = DC + CE$

Hence, $AB = DE$

SHORT ANSWER Type-II Questions (SA-II)

[3 marks]

7. In the given figure, $AM = \frac{1}{2} AB$ and $AN = \frac{1}{2} AC$.



Show that $AB = AC$.

Ans.

$$AM = \frac{1}{2} AB$$
$$AN = \frac{1}{2} AC$$

Now, $AM = \frac{1}{2} AB$

$$2 = \frac{AB}{AM} \quad \dots(i)$$

$$AN = \frac{1}{2} AC$$

$$2 = \frac{AC}{AN} \quad \dots(ii)$$

From eqs. (i) and (ii)

$$\frac{AC}{AN} = \frac{AB}{AM}$$

$$\frac{AC}{AN} = \frac{AB}{AM}$$

$$[\therefore AM = AN]$$

$$AC = AB$$

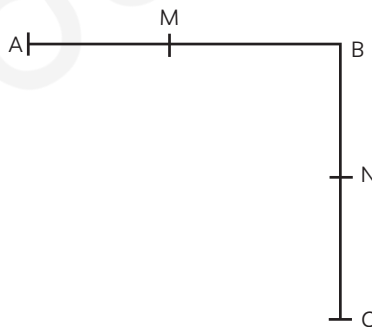
Hence, proved

LONG ANSWER Type Question (LA)

[4 & 5 marks]

8. (A) $AB = BC$, M is the mid-point of AB and N is the mid-point of BC. Show that $AM = NC$.

(B) $BM = BN$, M is the mid-point of AB and N is the mid-point of BC. Show that $AB = BC$.



Ans. (A) Given: $AB = BC$, M is the mid-point of AB and N is the mid-point of BC.

As M is the mid-point of AB, $AM = MB = \frac{1}{2} AB$.

As N is the mid-point of BC, $BN = NC = \frac{1}{2} BC$.

According to Euclid's seventh axiom, things which are halves of the same things are equal to one another.

$$\Rightarrow MA = NC$$



$$\therefore AM = NC$$

(B) Given: $BM = BN$, M is the mid-point of AB and N is the mid-point of BC.
Given that $BM = BN$.

As M is the mid-point of AB, $AM = MB = \frac{1}{2} AB$

As N is the mid-point of BC, $BN = NC = \frac{1}{2} BC$

According to Euclid's sixth axiom, things which are double of the same thing are equal to one another.

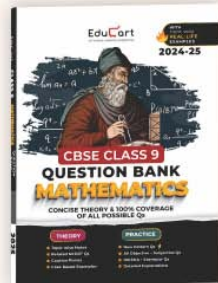
$$\Rightarrow AB = BC$$

Therefore, $AB = BC$.



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